

Do not use the STL <list> or Templates in your code. Do not change the specification or member functions interfaces or add extra member variables, if you do you will get zero marks

1. Implement a **class** called `NodeofInt` class which is needed to implement the list class below. This class has only the following:

- two private member variables: `int theValue` and `NodeofInt* next`
- a constructor which sets the int value to 0 and the pointer to `NullPtr` and

2. Implement a **class** called `ListOfInts`.

This class should use a singly linked list to maintain a list of int values.

It will have only one private member variable `NodeofInt* head`

and the following methods:

- a constructor `ListOfInts()` and a destructor `~ListOfInts` methods
- `insertBack(int)` which will be passed an int value to insert at the back of the list
- `displayList()` which will print out all the data in the list
- `deleteMostRecent()` which will return the value of the most recently added int, and deletes it from the list.

Note: you should have 2 separate .h files, `ListOfInts.h` and `NodeofInt.h` and 2 separate .cpp files, `ListOfInts.cpp` and `NodeofInt.cpp`

3. Once you have parts above completed, implement a method `deleteInt(int pos)` which deletes the int found in the `pos` position on the list (provided the list has that many entries, otherwise return 0) This method should traverse the list to reach the position, return the value it finds and delete it. You will need to draw some diagrams first to decide how to manipulate the pointers (and you will need to maintain pointers to more than one `NodeofInt`

4. Implement a main function (in a separate `main.cpp` file) to test your List class functionality, this should create a List and call or invoke all its member functions.